

Scalability & Future Plan

Date	5 October 2026
Team ID	6ab2951f1266c775575a6b80
Project Name	Farm Management System
Maximum Marks	5 Marks

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Scalability: Solution cannot handle data from more than 100 sensors simultaneously.	Data loss or significant latency for larger farms.	High
2	Offline Access: System requires a constant internet connection.	Farm activities might be logged late or missed.	High
3	Hardware Cost: Sensors use proprietary wireless protocol, which is expensive.	Barriers to entry for smaller farm operations.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade
1	User Load	Single user login.	Implement RBAC (Role-Based Access Control) for multiple concurrent users (e.g., Farmers, Agri-Analyst, Labor).
2	Data Storage	Simple SQLite database.	Migrate to PostGIS enabled PostgreSQL with read/write splitting.
3	Performance	Serial data processing.	Use asynchronous task queues (e.g., Celery) for data ingesting.
4	Security	Basic password protection.	Implement OAuth 2.0 and API rate limiting.

Future Roadmap:

Phase	Planned Feature / Enhancement		Target Timeline	Expected Impact
Phase 1	Localized Sensor-to-Cloud Integration	Develop a low-cost IoT gateway.	6 months	50% reduction in deployment cost.
Phase 2	Mobile App for Farmers	Develop an intuitive Android/iOS app.	9 months	Higher adoption and real-time alerts.
Phase 3	AI-based Crop Health Prediction	Use machine learning models for yield forecasting.	12 months	Improved farm decision making.
Phase 4	Market Price API Integration	Connect to global agri-commodity markets.	18 months	Better selling price insights for farmers.